Q. 1 (Fun with distributions)**.** Let $X \sim \text{Expon}(\lambda)$ and $Z = 1 - e^{-\lambda X}$. Note that X is a nonnegative random variable, and Z is a random variable with values in $[0, 1[$.

a. What is the expectation of the random variable Z?

b. What is the density of Z?

Now let X be a random variable with values in $[1, \infty[$ such that

$$\mathbf{P}\{X \in dx\} = cx^{-4}dx,$$

for some constant c that must be chosen properly. This is an example of a *Pareto distribution*, which can be used to model anything from the wealth of a typical person in the United States to the sizes of cities, grains of sand, and rainfalls.

c. What is the correct constant c?

d. Compute $\mathbf{E}(X)$, $\text{Var}(X)$, and $\mathbf{E}(X^3)$.

e. Let $Z = 1 - X^{-3}$. What is the density of Z?

Let X be any random variable that has a continuous and strictly increasing cumulative distribution function $F(x) = \mathbf{P}\{X \leq x\}$. Define the random variable $Z = F(X)$.

f. What is the cumulative distribution function of the random variable Z? Use your answer to explain the answers you found in parts b and e.

Q. 2 (Sibling rivalry)**.** You have two children. Their heights X_1, X_2 are independent and exponentially distributed with mean 5.5ft.

a. What is the probability that both children have the same height?

b. What is the probability that the first child is more than twice as tall as the second?

c. What is the probability that the *tallest* of the two children is more than twice as tall as the *shortest* of the two children?

Q. 3 (Men and women)**.** The height of adult men has an exponential distribution with mean $5\frac{3}{4}$ feet, while the height of adult women has an exponential distribution with mean $5\frac{1}{3}$ feet. In the US, 51% of people are women.

a. What is the average height of a person in the US?

b. Let X be the height of a random person in the US. What is the density of X?

c. You are told by a psychic that a mystery person will visit you shortly who is precisely $5\frac{1}{2}$ feet tall. What is the conditional probability that this person is a woman?

Q. 4 (Conditioning on sums)**.** Suppose that X and Y are independent random variables and that each is uniformly distributed in the interval $[0, 1]$.

a. What is the density of $X + Y$?

b. Suppose you are not told the individual outcomes of X, Y, but only their sum. What is the expected value of X, given that $X + Y = z$ (for $z \in \mathbb{R}$)?

Q. 5 (Telemarketers)**.** Every morning, a telemarketer gets a list of N people he must call during the day. We assume that N is Poisson distributed with mean 30.

The telemarketer does not want to be on the phone long into the night, as that would cut into his time playing video games. So, he adjusts his calling behavior according to the number of calls he has to make. When he has many calls to make, he tries to make each call relatively short so he can get done in a reasonable time. If he has only few calls to make, then he can take his time for each call so he can be more persuasive. If the number of calls he has to make is $N = n$, then the duration of each call will be exponentially distributed with parameter n. What is the expected amount of time it takes for the telemarketer to complete all his calls for the day?